THE TWISTED INVERSE IMAGE OF A NORMALIZED DUALIZING COMPLEX ALONG A PROPER MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0AUD (sources/stacks-duality/pinned-original-source.tex line 5541).

Allowed standard prerequisites (editorial).

- Composition of upper shrieks. Same chapter, Lemma labelled lemma-upper-shriek-composition, cited at proof lines 5553-5557. Gives $f^! \circ q^! = p^!$ for the separated case.
- Good dualizing complexes over both source and target. Same chapter, Lemma labelled lemma-good-over-both, cited at proof lines 5557-5559. Reduces the general case to $Y = S$.
- Normalized dualizing complexes. Same chapter, Situation labelled situation-dualizing, cited at proof lines 5540-5549. Supplies the normalization making the shriek formulae apply.

Source reference. Original reference 0AU4: sources/stacks-duality/pinned-original-source.tex, source lines 5259–5259. Same-chapter pointer into the author environment situation at source line 5259; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A9E: sources/stacks-duality/pinned-original-source.tex, source lines 653–653. Same-chapter pointer into the author environment lemma at source line 653; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0ATX: sources/stacks-duality/pinned-original-source.tex, source lines 4161–4161. Same-chapter pointer into the author environment lemma at source line 4161; invoked at proof line 5556 of the counted proof.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AUB: sources/stacks-duality/pinned-original-source.tex, source lines 5487–5487. Same-chapter pointer into the author environment lemma at source line 5487; invoked at proof line 5557 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $(S, \omega_S^\bullet)$ be as in Situation 0AU4. Let $f : X \to Y$ be a proper morphism of schemes of finite type over S. Let $\omega_X^\bullet$ and $\omega_Y^\bullet$ be dualizing complexes normalized relative to $\omega_S^\bullet$. Let a be the right adjoint of Lemma 0A9E for f. Then there is a canonical isomorphism $a(\omega_Y^\bullet) = \omega_X^\bullet$.*

Proof. Let $p : X \to S$ and $q : Y \to S$ be the structure morphisms. If X and Y are separated over S, then this follows from the fact that $\omega_X^\bullet = p^!\omega_S^\bullet$, $\omega_Y^\bullet = q^!\omega_S^\bullet$, $f^! = a$, and $f^! \circ q^! = p^!$ (Lemma 0ATX). In the general case we first use Lemma 0AUB to reduce to the case $Y = S$. In this case X and Y are separated over S and we've just seen the result. □